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Radar-based target tracker

Abstract

In an embodiment, a method includes: receiving a range-Doppler image (RDI) based on raw data from a radar sensor; performing moving target indication (MTI) filtering on the RDI to generate a first filtered radar image; performing constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image; performing minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI); performing CFAR detection on the RAI to generate a third filtered radar image; generating a point set based on the third filtered radar image; clustering targets of the point set; and tracking at least one of the clustered targets using a Kalman filter.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to an electronic system and method, and, in particular embodiments, to a radar-based target tracker.

BACKGROUND

(2) Applications in the millimeter-wave frequency regime have gained significant interest in the past few years due to the rapid advancement in low cost semiconductor technologies, such as silicon germanium (SiGe) and fine geometry complementary metal-oxide semiconductor (CMOS) processes. Availability of high-speed bipolar and metal-oxide semiconductor (MOS) transistors has led to a growing demand for integrated circuits for millimeter-wave applications at e.g., 24 GHz, 60 GHz, 77 GHz, and 80 GHz and also beyond 100 GHz. Such applications include, for example, automotive radar systems and multi-gigabit communication systems.

(3) In some radar systems, the distance between the radar and a target is determined by transmitting a frequency modulated signal, receiving a reflection of the frequency modulated signal (also referred to as the echo), and determining a distance based on a time delay and/or frequency difference between the transmission and reception of the frequency modulated signal. Accordingly, some radar systems include a transmitting antenna for transmitting the radio-frequency (RF) signal, and a receiving antenna for receiving the reflected RF signal, as well as the associated RF circuits used to generate the transmitted signal and to receive the RF signal. In some radar systems, multiple antennas may be used to implement directional beams using phased array techniques. A multiple-input and multiple-output (MIMO) configuration with multiple chipsets can be used to perform coherent and non-coherent signal processing.

SUMMARY

(4) In accordance with an embodiment, a method includes: receiving a range-Doppler image (RDI) based on raw data from a radar sensor; performing moving target indication (MTI) filtering on the RDI to generate a first filtered radar image; performing constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image; performing minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI); performing CFAR detection on the RAI to generate a third filtered radar image; generating a point set based on the third filtered radar image; clustering targets of the point set; and tracking at least one of the clustered targets using a Kalman filter.

(5) In accordance with an embodiment, a millimeter-wave radar system includes: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals; and a processing system configured to: generate a range-Doppler image (RDI) based on the reflected radar signals, perform moving target indication (MTI) filtering on the RDI to generate a first filtered radar image, perform constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image, perform minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI), perform CFAR detection on the RAI to generate a third filtered radar image, generate a point set based on the third filtered radar image, cluster targets of the point set, and track at least one of the clustered targets using a Kalman filter.

(6) In accordance with an embodiment, a device includes: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals; a communication device including a transmitter configured to radiate power with a first intensity; and a processing system configured to: generate a range-Doppler image (RDI) based on the reflected radar signals, perform

moving target indication (MTI) filtering on the RDI to generate a first filtered radar image, perform constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image, perform minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI), perform CFAR detection on the RAI to generate a third filtered radar image, generate a point set based on the third filtered radar image, cluster targets of the point set, track at least one of the clustered targets, and when a first target of the tracked clustered targets is detected within a predetermined distance from the transmitter of the communication device, cause a reduction of the radiated power from the first intensity to a second intensity lower than the first intensity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) For a more complete understanding of the present invention, and the advantages thereof, reference is now made to the following descriptions taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 shows a schematic diagram of a millimeter-wave radar system, according to an embodiment of the present invention;
- (3) FIG. 2 shows a sequence of chirps transmitted by the transmitter antenna of FIG. 1, according to an embodiment of the present invention;
- (4) FIG. 3 shows a flow chart of an embodiment method for detecting and tracking short-range targets, according to an embodiment of the present invention;
- (5) FIG. 4 shows a flow chart of an embodiment method for controlling the intensity of power radiated by a transmitter, according to an embodiment of the present invention;
- (6) FIGS. 5A and 5B illustrate various views of a laptop, according to an embodiment of the present invention;
- (7) FIG. 6 illustrates mobile device 600, according to an embodiment of the present invention;
- (8) FIG. 7 shows a flow chart of an embodiment method for detecting the virtual actuation of a button and, in response, triggering a predetermined action, according to an embodiment of the present invention;
- (9) FIGS. 8A-8D illustrate various views of a cooking system, according to an embodiment of the present invention;
- (10) FIG. 9 shows a flow chart of an embodiment method for detecting the presence of a target near a predetermined location, according to an embodiment of the present invention; and
- (11) FIGS. 10A and 10B illustrate various views of a laptop, according to an embodiment of the present invention.
- (12) Corresponding numerals and symbols in different figures generally refer to corresponding parts unless otherwise indicated. The figures are drawn to clearly illustrate the relevant aspects of the preferred embodiments and are not necessarily drawn to scale.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(13) The making and using of the embodiments disclosed are discussed in detail below. It should be appreciated, however, that the present invention provides many applicable inventive concepts that can be embodied in a wide variety of specific contexts. The specific embodiments discussed are merely illustrative of specific ways to make and use the invention, and do not limit the scope of the invention.

(14) The description below illustrates the various specific details to provide an in-depth understanding of several example embodiments according to the description. The embodiments may be obtained without one or more of the specific details, or with other methods, components, materials and the like. In other cases, known structures, materials or operations are not shown or

described in detail so as not to obscure the different aspects of the embodiments. References to “an embodiment” in this description indicate that a particular configuration, structure or feature described in relation to the embodiment is included in at least one embodiment. Consequently, phrases such as “in one embodiment” that may appear at different points of the present description do not necessarily refer exactly to the same embodiment. Furthermore, specific formations, structures or features may be combined in any appropriate manner in one or more embodiments.

(15) Embodiments of the present invention will be described in specific contexts, e.g., a radar-based short-range target tracker, which may be used in applications such as safety-related applications and human machine interface (HMI). Some embodiments may be used in other applications.

(16) In an embodiment of the present invention, a radar processing chain detects and tracks short-range targets. Static targets and clutter are removed by performing moving target indication (MTI) filtering on radar-Doppler images (RDIs). Constant false alarm rate (CFAR) is performed on the filtered RDIs before performing digital beamforming, which may advantageously reduce the computational complexity of the radar processing chain, which may allow for real-time operation.

(17) Embodiments of the present invention may be used in safety applications, such as by automatically causing the reduction of radiated power by a radio communication transmitter when a human (or a portion of a human) is near the radio communication transmitter and/or by warning a user when the user is too close to a predetermined location. Embodiments of the present invention may also be used in HMI applications, such as for detecting the actuation of virtual buttons.

(18) A radar, such as a millimeter-wave radar, may be used to detect and track one or more (e.g., short-range) targets. For example, FIG. 1 shows a schematic diagram of millimeter-wave radar system **100**, according to an embodiment of the present invention. Millimeter-wave radar system **100** includes millimeter-wave radar sensor **102** and processing system **104**.

(19) During normal operation, millimeter-wave radar sensor **102** operates as a frequency-modulated continuous-wave (FMCW) radar sensor and transmits a plurality of TX radar signals **106**, such as chirps, towards scene **120** using one or more transmitter (TX) antenna **114**. The radar signals **106** are generated using RF and analog circuits **130**. The radar signals **106** may be, e.g., in the 20 GHz to 122 GHz range. Other frequencies may also be used.

(20) In some embodiments, the field of view of millimeter-wave radar sensor **102** covering scene **120** has a short-range, such as below 1 meter, such as within 35 cm.

(21) The objects in scene **120** may include one or more static or moving objects, such as a surface (e.g., table, countertop, etc.), a keyboard, a wall, and a portion of a human, such as a hand, head, or finger. Other objects may also be present in scene **120**.

(22) The radar signals **106** are reflected by objects in scene **120**. The reflected radar signals **108**, which are also referred to as the echo signal, are received by a plurality of receiving (RX) antennas. RF and analog circuits **130** processes the received reflected radar signals **108** using, e.g., band-pass filters (BPFs), low-pass filters (LPFs), mixers, low-noise amplifier (LNA), and/or intermediate frequency (IF) amplifiers in ways known in the art to generate an analog signal $x_{\text{sub.outa}}(t)$ and $x_{\text{sub.outb}}(t)$.

(23) The analog signal $x_{\text{sub.outa}}(t)$ and $x_{\text{sub.outb}}(t)$ are converted to raw digital data $x_{\text{sub.out_dig}}(n)$ using analog-to-digital converter (ADC) **112**. The raw digital data $x_{\text{sub.out_dig}}(n)$ is processed by processing system **104** to detect one or more targets and their position. In some embodiments, processing system **104** may also be used to track one or more targets in scene **120**.

(24) Although FIG. 1 illustrates a radar system with two receiver antennas **116** (antennas **116a** and **116b**), it is understood that more than two receiver antennas **116**, such as three or more, may also be used. In some embodiments, using more receiver antennas **116** may result in higher spatial resolution.

(25) Although FIG. 1 illustrates a radar system with a single transmitter antenna **114**, it is

understood that more than one transmitter antenna **114**, such as two or more, may also be used.

(26) Controller **100** controls one or more circuits of millimeter-wave radar sensor **102**, such as RF and analog circuit **130** and/or ADC **112**. Controller **100** may be implemented, e.g., as a custom digital or mixed signal circuit, for example. Controller **100** may also be implemented in other ways, such as using a general-purpose processor or controller, for example. In some embodiments, processing system **104** implements a portion or all of controller **100**.

(27) Processing system **104** may be implemented with a general-purpose processor, controller or digital signal processor (DSP) that includes, for example, combinatorial circuits coupled to a memory. In some embodiments, processing system **104** may be implemented as an application specific integrated circuit (ASIC). In some embodiments, processing system **104** may be implemented with an ARM, RISC, or x86 architecture, for example. In some embodiments, processing system **104** may include an artificial intelligence (AI) accelerator. Some embodiments may use a combination of hardware accelerator and software running on a DSP or general-purpose microcontroller. Other implementations are also possible.

(28) In some embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented inside the same integrated circuit (IC). For example, in some embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented in respective semiconductor substrates that are integrated in the same package. In other embodiments, millimeter-wave radar sensor **102** and a portion or all of processing system **104** may be implemented in the same monolithic semiconductor substrate. In some embodiments, millimeter-wave radar sensor **102** and processing system **104** are implemented in respective integrated circuits. In some embodiments, a plurality of integrated circuits is used to implement millimeter-wave radar sensor **102**. In some embodiments, a plurality of integrated circuits is used to implement processing system **104**. Other implementations are also possible.

(29) As a non-limiting example, RF and analog circuits **130** may be implemented, e.g., as shown in FIG. **1**. During normal operation, voltage-controlled oscillator (VCO) **136** generates radar signals, such as a linear frequency chirps (e.g., from 57 GHz to 64 GHz, or from 76 GHz to 77 GHz), which are transmitted by transmitting antenna **114**. The VCO **136** is controlled by PLL **134**, which receives a reference clock signal (e.g., 80 MHz) from reference oscillator **132**. PLL **134** is controlled by a loop that includes frequency divider **138** and amplifier **140**. Amplifier **137** may be used to drive transmitting antenna **114**.

(30) The TX radar signals **106** transmitted by transmitting antenna **114** are reflected by objects in scene **120** and received by receiving antennas **116a** and **116b**. The echo received by receiving antennas **116a** and **116b** are mixed with a replica of the signal transmitted by transmitting antenna **114** using mixer **146a** and **146b**, respectively, to produce respective intermediate frequency (IF) signals $x_{\text{sub}}.\text{IFa}(t)$ $x_{\text{sub}}.\text{IFb}(t)$ (also known as beat signals). In some embodiments, the beat signals $x_{\text{sub}}.\text{IFa}(t)$ and $x_{\text{sub}}.\text{IFb}(t)$ have a bandwidth between 10 kHz and 1 MHz. Beat signals with a bandwidth lower than 10 kHz or higher than 1 MHz is also possible. Amplifiers **145a** and **145b** may be used to receive the reflected radar signals from antennas **116a** and **116b**, respectively.

(31) Beat signals $x_{\text{sub}}.\text{IFa}(t)$ $x_{\text{sub}}.\text{IFb}(t)$ are filtered with respective low-pass filters (LPFs) **148a** and **148b** and then sampled by ADC **112**. ADC **112** is advantageously capable of sampling the filtered beat signals $x_{\text{sub}}.\text{outa}(t)$ $x_{\text{sub}}.\text{outb}(t)$ with a sampling frequency that is much smaller than the frequency of the signal received by receiving antennas **116a** and **116b**. Using FMCW radars, therefore, advantageously allows for a compact and low cost implementation of ADC **112**, in some embodiments.

(32) The raw digital data $x_{\text{sub}}.\text{out_dig}(n)$, which in some embodiments include the digitized version of the filtered beat signals $x_{\text{sub}}.\text{outa}(t)$ and $x_{\text{sub}}.\text{outb}(t)$, is (e.g., temporarily) stored, e.g., in matrices of $N_{\text{sub}}.\text{c} \times N_{\text{sub}}.\text{s}$ per receiving antenna **116**, where $N_{\text{sub}}.\text{c}$ is the number of chirps considered in a frame and $N_{\text{sub}}.\text{s}$ is the number of transmit samples per chirp, for further processing by processing system **104**.

(33) In some embodiments, ADC **112** is a 12-bit ADC with multiple inputs. ADCs with higher resolution, such as 14-bits or higher, or with lower resolution, such as 10-bits, or lower, may also be used. In some embodiments, an ADC per receiver antenna may be used. Other implementations are also possible.

(34) FIG. **2** shows a sequence of chirps **106** transmitted by TX antenna **114**, according to an embodiment of the present invention. As shown by FIG. **2**, chirps **106** are organized in a plurality of frames (also referred to as physical frames) and may be implemented as up-chirps. Some embodiments may use down-chirps or a combination of up-chirps and down-chirps, such as up-down chirps and down-up chirps. Other waveform shapes may also be used.

(35) As shown in FIG. **2**, each frame may include a plurality of chirps **106** (also referred to, generally, as pulses). For example, in some embodiments, the number of chirps in a frame is 64. Some embodiments may include more than 64 chirps per frame, such as 96 chirps, 128 chirps, 256 chirps, or more, or less than 64 chirps per frame, such as 32 chirps or less.

(36) In some embodiments, frames are repeated every FT time. In some embodiments, FT time is 50 ms. A different FT time may also be used, such as more than 50 ms, such as 60 ms, 100 ms, 200 ms, or more, or less than 50 ms, such as 45 ms, 40 ms, or less.

(37) In some embodiments, the FT time is selected such that the time between the beginning of the last chirp of frame n and the beginning of the first chirp of frame $n+1$ is equal to PRT. Other embodiments may use or result in a different timing.

(38) The time between chirps of a frame is generally referred to as pulse repetition time (PRT). In some embodiments, the PRT is 5 ms. A different PRT may also be used, such as less than 5 ms, such as 4 ms, 2 ms, 0.5 ms, or less, or more than 5 ms, such as 6 ms, or more.

(39) The duration of the chirp (from start to finish) is generally referred to as chirp time (CT). In some embodiments, the chirp time may be, e.g., 64 μ s. Higher chirp times, such as 128 μ s, or higher, may also be used. Lower chirp times, may also be used.

(40) In some embodiments, the chirp bandwidth may be, e.g., 4 GHz. Higher bandwidth, such as 6 GHz or higher, or lower bandwidth, such as 2 GHz, 1 GHz, or lower, may also be possible.

(41) In some embodiments, the sampling frequency of millimeter-wave radar sensor **102** may be, e.g., 1 MHz. Higher sampling frequencies, such as 2 MHz or higher, or lower sampling frequencies, such as 500 kHz or lower, may also be possible.

(42) In some embodiments, the number of samples used to generate a chirp may be, e.g., 64 samples. A higher number of samples, such as 128 samples, or higher, or a lower number of samples, such as 32 samples or lower, may also be used.

(43) FIG. **3** shows a flow chart of embodiment method **300** for detecting and tracking short-range targets, according to an embodiment of the present invention. Method **300** may be implemented by processing system **104**.

(44) During step **302**, a range fast Fourier transform (FFT) is performed on the raw digital data $x_{\text{sub.out_dig}}(n)$ (e.g., on data from each receiving channel, e.g., from each receiving antenna **116**). For example, in some embodiments, a windowed FFT having a length of a chirp (e.g., **106**) is calculated for each of a predetermined number of chirps (e.g., all chirps) in a frame. The result of the range FFT is an indication of energy distribution across ranges for each chirp.

(45) During step **304**, a Doppler FFT is performed on the out range FFT data (e.g., for each receiving antenna **116**) generated during step **302**. For example, in some embodiments, an FFT is calculated across each range bin over a number of consecutive periods to extract Doppler information. The result of step **304** are range Doppler maps (also known as range-Doppler images or RDIs) for each of the receiving channels (e.g., for each receiving antenna **116**).

(46) During step **306**, two-dimensional (2D) moving target indication (MTI) filtering is applied to each RDI to generate first filtered images. In some embodiments, performing MTI filtering serves to discriminate a target against the clutter. For example, in some embodiments, after MTI filtering is performed on the RDIs, only targets with high motion are retained as their energy varies across

Doppler images. Thus, in some embodiments, after 2D MTI filtering, a target (e.g., a hand) may be identifiable in the filtered RDIs while information about the background (e.g., keyboard, surface, etc.) may be partially or fully removed from the filtered RDIs. In some embodiments, the 2D MTI filter is implemented as a first order finite impulse response (FIR) filter.

(47) During step **308**, target detection is performed using constant false alarm rate (CFAR), such as ordered-statistic (OS) CFAR (OS-CFAR) on the filtered RDIs generated during step **306** to generated second filtered images (also referred to as RD masked detection images). During step **308**, the CFAR detector generates a detection image in which, e.g., “ones” represent targets and “zeros” represent non-targets based on the power levels of the RDI when compared with a, e.g., predetermined threshold. The RD detection image operates as a mask and is multiplied by the original RDI (output of step **306**) to obtain an RD masked image (at the output of step **308**) in which only details of the detected targets are non-zero.

(48) In some embodiments, detection of targets during step **308** is advantageously enabled by the relatively high signal-to-noise ratio (SNR) of moving targets due to the short-range field of view of millimeter-wave radar sensor **102**.

(49) In some embodiments, the performance of steps **306**, **308** and **310** advantageously allows for the detection of macro-motions as well as micro-motions of targets while ignoring non-targets (e.g., background, e.g., static, clutter).

(50) In some embodiments, performing target detecting on the RDIs before performing digital beamforming advantageously reduced computational complexity and allows for real-time operation.

(51) During step **310**, (e.g., 1D) digital beamforming (e.g., 1D Capon) is applied to estimate angle from the data across virtual channels from the antenna array configuration for each range that has passed step **308** to generate 2D range-angle image (RAI).

(52) During step **312**, target detection is performed using CFAR (e.g., OS-CFAR) on the RAI detection images generated during step **310** to generate third filtered images (also referred to as RA masked detection images). During step **312**, the CFAR detector generates a detection image in which, e.g., “ones” represent targets and “zeros” represent non-targets based on the power levels of the RAI when compared with a, e.g., predetermined threshold. The RA detection image operates as a mask and is multiplied by the original RAI (output of step **310**) to obtain an RA masked detection image (at the output of step **312**) in which only details of the detected targets are non-zero.

(53) During step **314**, a conversion from Polar coordinates to Cartesian coordinates is performed. In some embodiments, step **314** is omitted.

(54) During step **316**, a set of detection points is generated. In some embodiments, each detection point includes: 2D coordinates, a signal strength value (indicative of the signal strength of target located at the 2D coordinates), and a Doppler value (indicative of the level of movement of the target at the 2D coordinates). In some embodiments, the 2D coordinates are Cartesian coordinates (e.g., x, y) (e.g., if step **314** is performed). In some embodiments, the 2D coordinates are Polar coordinates (e.g., range and angle) (e.g., if step **314** is not performed).

(55) In some embodiments, such as in embodiments in which all receiving antennas are aligned in the same (e.g., horizontal) axis, each point of the set of detection points include 2D coordinates. In some embodiments, such as in embodiments in which the receiving antennas are arranged in an L shape, each point of the set of detection points include 3D coordinates.

(56) During step **318**, the points (targets) of the set of detection points are clustered, e.g., using a density-based spatial clustering of applications with noise (DBSCAN) algorithm to associate points (targets) to clusters. The output of DBSCAN is one or more groupings (clusters) of detected points, so that each grouping is associated with a respective target.

(57) In some embodiments, each clustered target is represented as a single target having a point location, which may be the centroid of the cluster. For example, in some embodiments, a human hand may appear as a collection of targets before step **318**. After step **318**, a single point (e.g.,

corresponding to a geometric center of the hand) is used to represent the human hand.

(58) In some embodiments, using Cartesian coordinates (e.g., by performing step **314**) advantageously allows for an easier (e.g., less computationally expensive) implementation of the clustering step (**318**) since Cartesian coordinates are advantageously linear irrespective of the range. For example, in some embodiments, the same threshold (e.g., clustering points within 6 cm) may be performed to points near millimeter-wave radar sensor **102** and to points far from millimeter-wave radar sensor **102**.

(59) During step **320**, the detected clusters are associated to respective tracks, e.g., using a probabilistic data association filter (PDAF) or nearest neighbor approach. In some embodiments, a (e.g., unscented) Kalman filter is used to filter and smoothen the tracks. In some embodiments, all features of the clusters (e.g., coordinates, signal strength value, and Doppler value) are tracked and filtered using the Kalman filter. In some embodiments, the Kalman filter may only track a subset of the features of the cluster (e.g., just the coordinates, such as (x,y)). The Kalman filter may be implemented in any way known in the art.

(60) During step **322**, visualization of the clusters (e.g., in a static manner, in real time, or over time) may be displayed, e.g., in a display.

(61) Compared to camera-based target detection and tracking, target detection and tracking based on radar measurements can have some advantages such as: invariant to illumination conditions and preserving privacy.

(62) Compared to Lidar-based solution target detection, target detection and tracking based on radar measurements can have some advantages such as being less expensive.

(63) In some embodiments, method **300** may be performed to detect the presence of a human or a portion of human near a radiation source and reduce the power emitted by the radiation source in response to the detection of the presence of the human or portion of the human near the radiation source (e.g., to reduce or minimize the chance of injury to the human or portion of the human). For example, radiated power of a 5G NR transmitter may be focused on a narrow zone, which can cause health issues due to a high specific absorption rate (SAR) level. It may be desirable for communication devices, such as communication devices operating according to the 5G NR communication standard, to maintain the radiated power level low when a human or a portion of a human is near the radiation source to keep the SAR level below acceptable levels (which may be dictated by governmental organizations, such as the federal communication commission (FCC), and other global safety guidelines).

(64) FIG. **4** shows a flow chart of embodiment method **400** for controlling the intensity of power radiated by a transmitter, according to an embodiment of the present invention. Method **400** may be implemented by processing system **104**.

(65) During step **402**, it is determined whether there is a target within the field of view (FoV) of millimeter-wave radar sensor **102**. If a target is detected, step **404** is performed. If a target is not detected, step **410** is performed.

(66) During step **404**, the location of the detected target (e.g., a hand) is obtained, e.g., by performing method **300**, with respect to the location of a radiation source. In some embodiments, the target location is obtained in Cartesian coordinates, which is advantageously linear with respect to the range from millimeter-wave radar sensor **102**. In some embodiments, more than one target is detected during step **404**.

(67) During step **406**, the distance between the location of each of the detected targets and the location of the source is determined. If the distance of at least one detected target is lower than a predetermined threshold (e.g., 6 cm), the power radiated by the radiation source is reduced during step **408**. If the distance is larger than the predetermined threshold, step **410** is performed.

(68) During step **410**, the radiated power of the radiation source is compared with a target (e.g., optimum) level. If the radiation power is below the target level (e.g., due to a reduction because of the performance of step **408**), the power radiated by the radiation source is increased (e.g., to the

target level) during step **412**.

(69) Method **400** may be implemented in applications having at least one radio transmitter, such as an 4G LTE radio transmitter, 5G NR radio transmitter, or other types of communication transmitters. For example, in some embodiment, method **400** may be implemented by a mobile device, such as a laptop, smartphone, tablet, etc. In some embodiments, method **400** may be implemented by a non-mobile device, such as a smart thermostat, a smart TV, etc.

(70) FIGS. **5A** and **5B** illustrate various views of laptop **500**, according to an embodiment of the present invention. Laptop **500** includes millimeter-wave radar system **100**, radio communication transmitter **504**, display **506**, bezel **514**, keyboard **508**, and hinges **510**. As shown in FIGS. **5A** and **5B**, radio communication transmitter **504** is within the field of view **502** of millimeter-wave radar sensor **102**.

(71) During normal operation, laptop **500** may perform method **400** (e.g., using processing system **104**). For example, when a target (e.g., a hand typing on keyboard **508**) is detected near radio communication transmitter **504** (output of step **406**=yes), the radiation power of radio communication transmitter **504** is reduced. By reducing the magnitude of power radiated by radio communication transmitter **504** when the target (e.g., the hand) is near radio communication transmitter **504**, some embodiments advantageously may prevent or reduce the risk of injury to the target (e.g., the hand) due to high radiation while maintaining optimum communication quality via radio communication transmitter **504** when the target is near radio communication transmitter **504**.

(72) In some embodiments, radio communication transmitter **504** operates according to the 5G NR standard. In some embodiments, radio communication transmitter **504** operates according to a communication protocol different from 5G NR, such as 4G LTE, or future generations of communication protocols, for example.

(73) In embodiments that include a plurality of radio communication transmitters **504**, a corresponding step **406** may be performed for each detected target.

(74) In some embodiments, such as in laptop **500**, all operational receiver antennas **116** are arranged along the x axis, which allows for the identification of 2D coordinates (e.g., x,y) for each target during step **316**. In some embodiments, some operational receiver antennas **116** are arranged along the x axis and some operational receiver antennas **116** are arranged along the z axis (such as 3 receiver antennas **116** arranged in L shape). Some such embodiments allow for the identification of 3D coordinates (e.g., x,y,z) for each target during step **316**.

(75) As shown in FIG. **5**, laptop **500** may implement millimeter-wave radar sensor **102** attached or inside hinge **510**. In some embodiments, other locations may be used, such as in bezel **514** (e.g., with field of view **502** designed to cover radio communication transmitter **504** during normal operation), for example.

(76) In some embodiments implementing method **400**, millimeter-wave radar sensor **102** may be collocated in the same location as radio communication transmitter **504**. For example, FIG. **6** illustrates mobile device **600**, according to an embodiment of the present invention. Mobile device **600** includes display **606**, bezel **614**, millimeter-wave radar system **100**, and radio communication transmitter **504**.

(77) As shown in FIG. **6**, radio communication transmitter **504** and millimeter-wave radar sensor **102** are collocated in the same location in bezel **614**. As also shown in FIG. **6**, field of view **604** of radiation power of radio communication transmitter **504** (e.g., substantially) overlaps with field of view **502** of millimeter-wave radar sensor **102** in the short-range region (and may have substantially identical angle).

(78) During normal operation, mobile device **600** may perform method **400** (e.g., using processing system **104**). For example, since fields of view **502** and **604** (e.g., substantially) overlap in the short range (e.g., at a range equal to the predetermined range, or lower), a target detected in field of view **502** near millimeter-wave radar sensor **102** (at short range) is also near radio communication transmitter **504**. Thus, in some embodiments, when a target (e.g., a hand) is detected in the field of

view **502** at a range equal to a predetermined range or lower (output of step **406**=yes), the radiation power of radio communication transmitter **504** is reduced. By reducing the magnitude of power radiated by radio communication transmitter **504** when the target (e.g., the hand) is near radio communication transmitter **504**, some embodiments advantageously may prevent or reduce the risk of injury to the target (e.g., the hand) due to high radiation while maintaining optimum communication quality via radio communication transmitter **504** when the target is near radio communication transmitter **504**.

(79) In embodiments that include a plurality of radio communication transmitters **504**, a corresponding plurality of millimeter-wave radar sensors **102** may be used to perform respective method **400**.

(80) In some embodiments, method **300** may be used for human machine interface (HMI). For example, FIG. 7 shows a flow chart of embodiment method **700** for detecting the virtual actuation of a button and, in response, triggering a predetermined action, according to an embodiment of the present invention. Method **700** may be implemented by processing system **104**.

(81) Method **700** includes steps **702**, **404**, **706** and **708**. Step **404** may be implemented in a similar manner as described with respect to FIG. 4.

(82) During step **702**, it is determined whether there is a target within the field of view (FoV) of millimeter-wave radar sensor **102**, where the field of view of millimeter-wave radar sensor covers the location of a virtual button. If a target is detected, after obtaining the location of the target during step **404**, the target location is compared with the location of the virtual button during step **706**. If the target is at the location of the virtual button for longer than a predetermined time (e.g., 2 seconds), a predetermined action is triggered during step **708**.

(83) In some embodiments, the predetermined action triggered during step **708** includes turning on or off a device or function or changing a parameter of a device (e.g., volume, intensity, etc.), for example.

(84) In embodiments having multiple virtual buttons, step **706** may be performed for the locations of each of the virtual buttons.

(85) FIGS. **8A** and **8B** illustrate various views of cooking system **800**, according to an embodiment of the present invention. Cooking system **800** includes hood **802**, projector **804**, and stove **806**. Stove **806** includes top surface **808** and burners **810**, **812**, **814** and **816**.

(86) FIGS. **8C** and **8D** illustrate a top view of stove **806** during operation, according to an embodiment of the present invention.

(87) As shown in FIGS. **8A** and **8B**, in an off mode of operation, virtual buttons are not displayed on top surface **808**. As shown in FIGS. **8C** and **8D**, in an on mode of operation, projector **804** displays one or more buttons (e.g., **820**, **822**, **824**, and **826**) on top surface **808**. During the on mode of operation, cooking system **800** may perform method **700** (e.g., using processing system **104**). For example, as illustrated in FIG. **8D**, when hand/finger **840** is detect over button **824** for longer than a predetermined time (output of step **706**=yes), the corresponding burner **814** is turned on.

(88) As shown in FIGS. **8A-8D**, in some embodiments, millimeter-wave radar sensor **102** may be disposed on surface **808**. In some embodiments, millimeter-wave radar sensor **102** may be disposed in other places, such as collocated with projector **804**, for example.

(89) In some embodiments, method **300** may be used to warn or take action when a target is too close to a predetermined location. For example, FIG. 9 shows a flow chart of embodiment method **900** for detecting the presence of a target near a predetermined location, according to an embodiment of the present invention. Method **900** may be implemented by processing system **104**.

(90) Method **900** includes steps **702**, **404**, **906** and **908**. Step **404** may be implemented in a similar manner as described with respect to FIG. 4. Step **702** may be implemented in a similar manner as described with respect to FIG. 7.

(91) If a target is detected during step **702**, after obtaining the location of the target during step **404**, the target location is compared with a predetermined location (e.g., the location of millimeter-wave

radar sensor **102**) during step **906**. If the target closer than a predetermined distance to the predetermined location, a predetermined action (e.g., warn a user that it is too close to the predetermined location) is triggered during step **908**.

(92) FIGS. **10A** and **10B** illustrate various views of laptop **1000**, according to an embodiment of the present invention. Laptop **1000** includes millimeter-wave radar system **100**, display **506**, bezel **514**, keyboard **508**, and hinges **510**.

(93) During normal operation, laptop **1000** may perform method **900** (e.g., using processing system **104**). For example, as illustrated in FIG. **10B**, head **1002** of a human may be within field of view **502**. When head **1002** is within the field of view **502** (output of step **702**=yes) if the location of the head is too close to display **506** (e.g., when the range from millimeter-wave radar sensor **102** is lower than a predetermined threshold), an action is taken (**908**), such as warning the user (e.g., via display **506**, using a sound, and/or in another manner) that the user is too close to display **506** (e.g., which may be harmful to the human eyes). In some embodiments, the brightness of display **506** may be automatically reduced when the target (e.g., head **1002**) is too close to display **506** (such brightness may automatically return to the default brightness once the target is no longer too close to display **506**).

(94) Example embodiments of the present invention are summarized here. Other embodiments can also be understood from the entirety of the specification and the claims filed herein.

(95) Example 1. A method including: receiving a range-Doppler image (RDI) based on raw data from a radar sensor; performing moving target indication (MTI) filtering on the RDI to generate a first filtered radar image; performing constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image; performing minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI); performing CFAR detection on the RAI to generate a third filtered radar image; generating a point set based on the third filtered radar image; clustering targets of the point set; and tracking at least one of the clustered targets using a Kalman filter.

(96) Example 2. The method of example 1, where each point of the point set includes a signal strength component and a Doppler component.

(97) Example 3. The method of one of examples 1 or 2, where each point of the point set includes Cartesian coordinates.

(98) Example 4. The method of one of examples 1 to 3, where cluster the targets includes clustering the targets based on the Cartesian coordinates of each point of the point set.

(99) Example 5. The method of one of examples 1 to 4, where each point of the point set includes Polar coordinates.

(100) Example 6. The method of one of examples 1 to 5, further including: transmitting radar signals using the radar sensor; receiving reflected radar signals using the radar sensor; generating the raw data based on the reflected radar signals; and generating the range-Doppler image based on the raw data.

(101) Example 7. The method of one of examples 1 to 6, where generating the range-Doppler image includes performing a range fast Fourier transform (FFT) on the raw data.

(102) Example 8. The method of one of examples 1 to 7, where the radar sensor includes two receiving antennas.

(103) Example 9. The method of one of examples 1 to 8, where the radar sensor includes more than two receiving antennas.

(104) Example 10. The method of one of examples 1 to 9, further including: radiating power with a first intensity using a transmitter of a communication device; and when a first target of the tracked clustered targets is detected within a predetermined distance from the transmitter of the communication device, reducing the radiated power from the first intensity to a second intensity lower than the first intensity.

(105) Example 11. The method of one of examples 1 to 10, where the radar sensor and the

transmitter of the communication device are collocated in the same location.

(106) Example 12. The method of one of examples 1 to 11, where a field of view of the radar sensor and a field of view of the transmitter of the communication device have a substantially identical angle.

(107) Example 13. The method of one of examples 1 to 12, where a field of view of the radar sensor covers the transmitter of the communication device.

(108) Example 14. The method of one of examples 1 to 13, where the communication device operates according to the 5G NR communication standard.

(109) Example 15. The method of one of examples 1 to 14, where the transmitter of the communication device is located adjacent to a keyboard of a laptop, and where the radar sensor is located in a bezel of a display of the laptop.

(110) Example 16. The method of one of examples 1 to 15, further including: projecting a button into a first surface; and when a first target of the tracked clustered targets is detected within a predetermined distance from the projected button, causing a first action to be performed.

(111) Example 17. The method of one of examples 1 to 16, where projecting the button includes projecting the button with a projector located in a hood of a kitchen.

(112) Example 18. The method of one of examples 1 to 17, where the radar sensor is collocated with the projector.

(113) Example 19. The method of one of examples 1 to 18, where the radar sensor is located in the first surface.

(114) Example 20. The method of one of examples 1 to 19, where the radar sensor is located in a bezel of a display, the method further including, when a first target of the tracked clustered targets is detected within a predetermined distance from the display, issue a flag indicative of a user being too close to the display.

(115) Example 21. The method of one of examples 1 to 20, where a first target of the tracked clustered targets corresponds to a center of a hand of a human.

(116) Example 22. A millimeter-wave radar system including: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals; and a processing system configured to: generate a range-Doppler image (RDI) based on the reflected radar signals, perform moving target indication (MTI) filtering on the RDI to generate a first filtered radar image, perform constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image, perform minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI), perform CFAR detection on the RAI to generate a third filtered radar image, generate a point set based on the third filtered radar image, cluster targets of the point set, and track at least one of the clustered targets using a Kalman filter.

(117) Example 23. A device including: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals; a communication device including a transmitter configured to radiate power with a first intensity; and a processing system configured to: generate a range-Doppler image (RDI) based on the reflected radar signals, perform moving target indication (MTI) filtering on the RDI to generate a first filtered radar image, perform constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image, perform minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI), perform CFAR detection on the RAI to generate a third filtered radar image, generate a point set based on the third filtered radar image, cluster targets of the point set, track at least one of the clustered targets, and when a first target of the tracked clustered targets is detected within a predetermined distance from the transmitter of the communication device, cause a reduction of the radiated power from the first intensity to a second intensity lower than the first intensity.

(118) While this invention has been described with reference to illustrative embodiments, this

description is not intended to be construed in a limiting sense. Various modifications and combinations of the illustrative embodiments, as well as other embodiments of the invention, will be apparent to persons skilled in the art upon reference to the description. It is therefore intended that the appended claims encompass any such modifications or embodiments.

Claims

1. A method comprising: receiving a range-Doppler image (RDI) based on raw data from a radar sensor; performing moving target indication (MTI) filtering on the RDI to generate a first filtered radar image; performing constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image; performing minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI); performing CFAR detection on the RAI to generate a third filtered radar image; generating a point set based on the third filtered radar image; clustering targets of the point set; and tracking at least one of the clustered targets using a Kalman filter.
2. The method of claim 1, wherein each point of the point set comprises a signal strength component and a Doppler component.
3. The method of claim 2, wherein each point of the point set comprises Cartesian coordinates.
4. The method of claim 3, wherein clustering the targets comprises clustering the targets based on the Cartesian coordinates of each point of the point set.
5. The method of claim 2, wherein each point of the point set comprises Polar coordinates.
6. The method of claim 1, further comprising: transmitting radar signals using the radar sensor; receiving reflected radar signals using the radar sensor; generating the raw data based on the reflected radar signals; and generating the range-Doppler image based on the raw data.
7. The method of claim 6, wherein generating the range-Doppler image comprises performing a range fast Fourier transform (FFT) on the raw data.
8. The method of claim 1, wherein the radar sensor comprises two receiving antennas.
9. The method of claim 1, wherein the radar sensor comprises more than two receiving antennas.
10. The method of claim 1, further comprising: radiating power with a first intensity using a transmitter of a communication device; and when a first target of the tracked clustered targets is detected within a predetermined distance from the transmitter of the communication device, reducing the radiated power from the first intensity to a second intensity lower than the first intensity.
11. The method of claim 10, wherein the radar sensor and the transmitter of the communication device are collocated in a same location.
12. The method of claim 11, wherein a field of view of the radar sensor and a field of view of the transmitter of the communication device have a substantially identical angle.
13. The method of claim 10, wherein a field of view of the radar sensor covers the transmitter of the communication device.
14. The method of claim 10, wherein the communication device operates according to a 5G NR communication standard.
15. The method of claim 10, wherein the transmitter of the communication device is located adjacent to a keyboard of a laptop, and wherein the radar sensor is located in a bezel of a display of the laptop.
16. The method of claim 1, further comprising: projecting a button into a first surface; and when a first target of the tracked clustered targets is detected within a predetermined distance from the projected button, causing a first action to be performed.
17. The method of claim 16, wherein projecting the button comprises projecting the button with a projector located in a hood of a kitchen.
18. The method of claim 17, wherein the radar sensor is collocated with the projector.

19. The method of claim 17, wherein the radar sensor is located in the first surface.
20. The method of claim 1, wherein the radar sensor is located in a bezel of a display, and the method further comprises, when a first target of the tracked clustered targets is detected within a predetermined distance from the display, issuing a flag indicative of a user being too close to the display.
21. The method of claim 1, wherein a first target of the tracked clustered targets corresponds to a center of a hand of a human.
22. A millimeter-wave radar system comprising: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals; and a processing system configured to: generate a range-Doppler image (RDI) based on the reflected radar signals, perform moving target indication (MTI) filtering on the RDI to generate a first filtered radar image, perform constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image, perform minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI), perform CFAR detection on the RAI to generate a third filtered radar image, generate a point set based on the third filtered radar image, cluster targets of the point set, and track at least one of the clustered targets using a Kalman filter.
23. A device comprising: a millimeter-wave radar sensor configured to transmit radar signals and receive reflected radar signals; a communication device comprising a transmitter configured to radiate power with a first intensity; and a processing system configured to: generate a range-Doppler image (RDI) based on the reflected radar signals, perform moving target indication (MTI) filtering on the RDI to generate a first filtered radar image, perform constant false alarm rate (CFAR) detection on the first filtered radar image to generate a second filtered radar image, perform minimum variance distortionless response (MVDR) beamforming on the second filtered radar image to generate a range-angle image (RAI), perform CFAR detection on the RAI to generate a third filtered radar image, generate a point set based on the third filtered radar image, cluster targets of the point set, track at least one of the clustered targets, and when a first target of the tracked clustered targets is detected within a predetermined distance from the transmitter of the communication device, cause a reduction of the radiated power from the first intensity to a second intensity lower than the first intensity.
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